

IE4497/EE6497 Pattern Recognition and Deep Learning

Tay Wee Peng

Nanyang Technological University



1. Introduction

- 1 Course Admin
- 2 Overview
- 3 Probability Review
- 4 Some Practice Problems

Outline

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Learning Outcomes

1. Explain basic **pattern recognition** and **machine learning** concepts.
2. Describe **Bayesian inference**.
3. Explain **mixture models** and **EM algorithm**.
4. Describe **Markov models** and **hidden Markov models**.
5. Explain **sampling** and Markov chain Monte Carlo (**MCMC**) techniques.
6. Differentiate various algorithms and architectures of **neural networks** and deep learning.
7. Describe **training** deep network techniques.
8. Describe **generative models** and **self-supervised learning**.
9. Describe some **applications** of pattern recognition and machine learning.

Teaching team:

- Tay Wee Peng (wptay@ntu.edu.sg): Weeks 1 – 7
- Wang Lipo (elpwang@ntu.edu.sg): Weeks 8 – 13

Pre-requisites

- Related courses: IE4483/EE6483, IE4414, EE6406, EE6407.
- This course: **concepts and underlying theories**.
 - ▶ Mathematically interpretable: probabilistic approach.
 - ▶ Reasoning behind selected models.
- Math requirements:
 - ▶ Elementary calculus: taking derivatives, solving equations, finding max/min.
 - ▶ Basic probability theory: pmf, pdf, expectations, their interpretations.
- Both **theory and practice** are important. **Key: get involved mentally and your hands dirty implementing Python codes.**
- Practice problems are available at end of each week's lecture slides. Attempt the questions without looking at the **solutions** first!
- **Welcome timely feedback**: how can I help you learn better?

Grading for UG students taking IE4497

Component	Weighting
Final Examination	60%
Quiz 1	10%
Quiz 2	10%
Homework 1	5%
Homework 2	5%
Class Participation	10%

- No late submission of homeworks.
- Quiz 1, Quiz 2, Final Exam: you can bring ONE double-sided A4-size reference sheet with texts handwritten or typed on the A4 paper (no sticky notes/post-it notes on the reference sheet). Zero mark applies if no valid LOA/MC.

Grading for PG students taking EE6497

Component	Weighting
Final Examination	60%
Quiz 1	10%
Quiz 2	10%
Homework 1	5%
Homework 2	5%
Class Participation	5%
Survey Report	5%

- Additional homework questions and survey report mandatory for PG students.
- No late submission of homeworks.
- Quiz 1, Quiz 2, Final Exam: you can bring ONE double-sided A4-size reference sheet with texts handwritten or typed on the A4 paper (no sticky notes/post-it notes on the reference sheet). Zero mark applies if no valid LOA/MC.

Course Content

Week	Topic	CA (due in week)
1	Introduction, Probability Review	
2	Bayesian Inference	
3	Mixture Models and the EM Algorithm	
4	Markov Models and Hidden Markov Models	
5	Sampling	Homework 1
6	Markov Chain Monte Carlo (MCMC)	
7	Summary	Quiz 1 - Weeks 1 to 4
	Recess	
8	Neural Networks	
9	Deep Learning (CNN)	
10	Training Deep Networks	
11	Training Deep Networks	Homework 2
12	Deep Network Architectures	Quiz 2 - Weeks 8 to 11
13	Applications	Survey report (for PG only)

Class Participation: Wooclap

- Short questions to test your understanding during lecture.
- You will earn the full Class Participation marks if you attend and attempt the Wooclap questions in **9 out of 13** lectures. [Have to be captured via NTULearn.]

Course Content



 **Week 1: Introduction, Probability Review**
This week introduces the foundational concepts of pattern recognition and machine learning, emphasizing the importance of probability in these fields.

  1_Introduction.pdf

 **W Wooclap**
Wooclap is an interactive platform which makes courses more interactive and measure the comprehension of students using smartphones.

Acknowledgements

Textbook References

- [B07] Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007.
- [M12] Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, The MIT Press, 2012.
- [HTF09] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, Springer, 2009.
- Material is taken liberally from [B07] and [M12].
- Some Python codes are from [M12] (<https://github.com/probml/pyprobml>), with slight modifications.
- Some data taken from [HFT09].
- Note: Supplementary materials are provided for reference only. It is good to understand them but you will not be tested on them.

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Wooclap

Course Content



Week 1: Introduction, Probability Review

This week introduces the foundational concepts of pattern recognition and machine learning, emphasizing the importance of probability in these fields.



1_Introduction.pdf



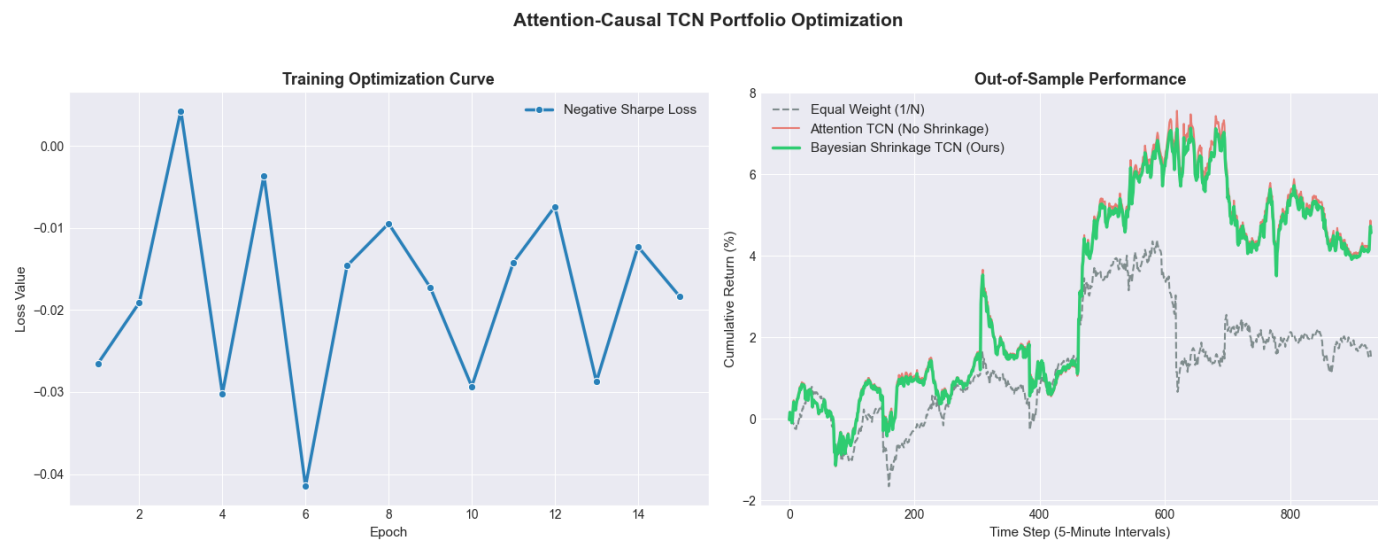
W Wooclap

Wooclap is an interactive platform which makes courses more interactive and measure the comprehension of students using smartphones.

Using Gemini Agent

What is the best PyTorch architecture for pattern recognition in financial time series to maximize the Sharpe ratio? Implement the model for optimizing a portfolio of tech stocks using intraday data from Yahoo Finance. Use out-of-sample backtesting to evaluate the performance of the model.

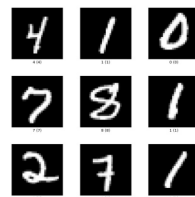
01_portfolio_optimization.ipynb



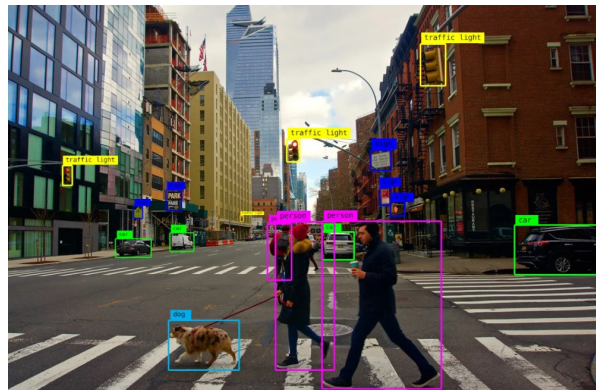
Machine Learning

- **Supervised learning:** learn a map $Y = f(X)$ to relate an input X to an output Y from a training set of input-output pairs $\mathcal{D} = \{(X_i, Y_i) : i = 1, \dots, n\}$.

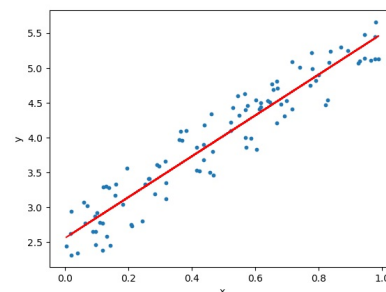
- ▶ Classification



- ▶ Object detection and recognition

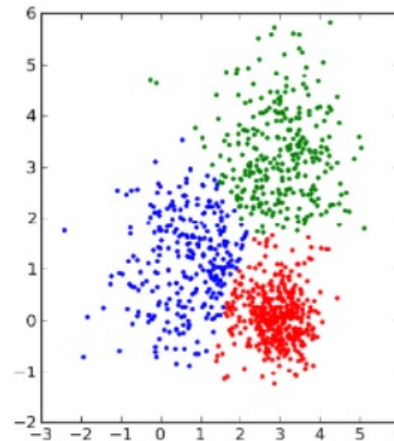


- ▶ Regression



Machine Learning

- **Unsupervised learning:** given only data $\mathcal{D} = \{X_i : i = 1, \dots, n\}$, want to discover structures or knowledge from the data.
 - ▶ Clustering



- ▶ Dimensionality reduction using latent factors



Machine Learning

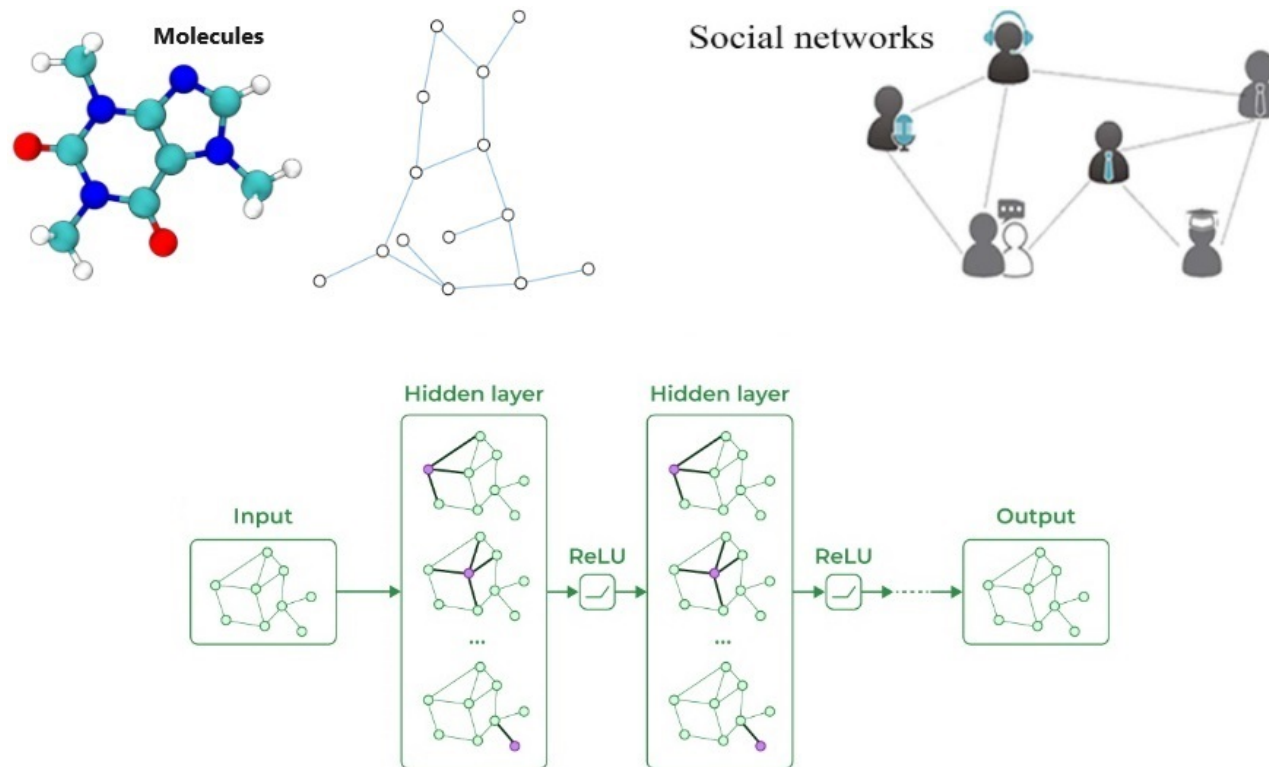
- **Unsupervised learning**: given only data $\mathcal{D} = \{X_i : i = 1, \dots, n\}$, want to discover structures or knowledge from the data.
 - ▶ Correlation and structure learning



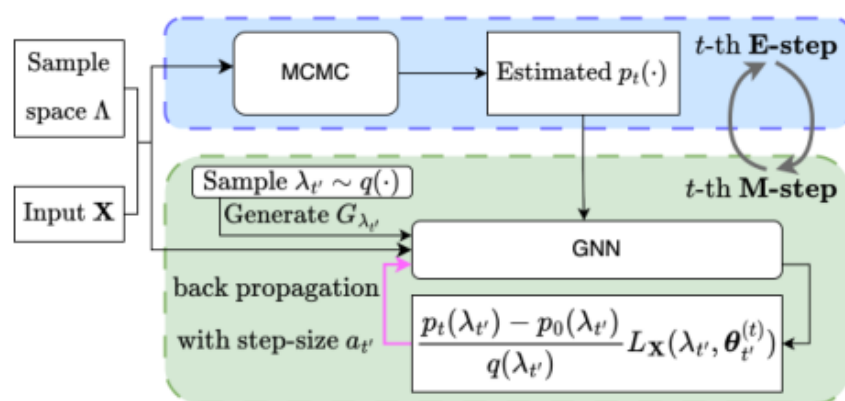
In the first half, we will learn how to rigorously perform pattern recognition/learning in a framework that is explainable most of the time.

Motivation: My Research

Graph neural networks: representation learning for unstructured graph-like data.



Motivation: My Research



EMGNN uses the following tools:

- Bayesian mixtures
- EM algorithm
- MCMC
- Graph neural networks

Table 3: Graph regression task on molecular datasets. Average test rmse reported, the lower the better.

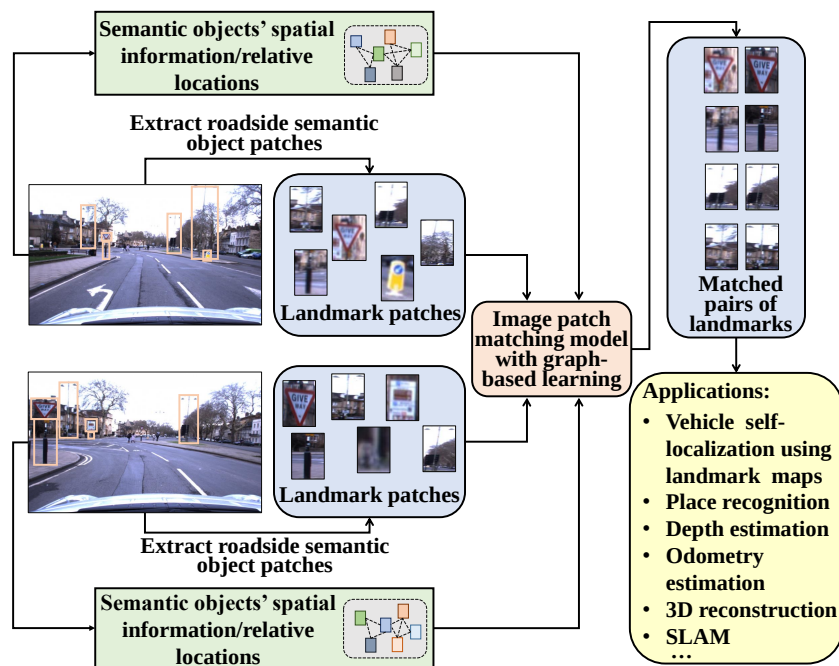
Datasets	Random split			Scaffold split		
	FreeSolv	ESOL	Lipophilicity	FreeSolv	ESOL	Lipophilicity
GCN	<u>1.157 $\pm$ 0.215</u>	<u>0.652 $\pm$ 0.073</u>	0.707 $\pm$ 0.030	<u>2.618 $\pm$ 0.298</u>	0.876 $\pm$ 0.037	0.760 $\pm$ 0.009
GAT	1.873 $\pm$ 0.522	0.837 $\pm$ 0.101	0.704 $\pm$ 0.058	2.942 $\pm$ 0.591	0.907 $\pm$ 0.034	0.777 $\pm$ 0.037
Weave	1.497 $\pm$ 0.251	0.798 $\pm$ 0.088	0.789 $\pm$ 0.059	3.129 $\pm$ 0.203	1.104 $\pm$ 0.063	0.844 $\pm$ 0.031
MPNN	1.388 $\pm$ 0.404	0.703 $\pm$ 0.075	<u>0.640 $\pm$ 0.025</u>	2.975 $\pm$ 0.775	1.117 $\pm$ 0.058	0.735 $\pm$ 0.019
AttentiveFP	1.275 $\pm$ 0.289	0.673 $\pm$ 0.085	0.719 $\pm$ 0.042	2.698 $\pm$ 0.297	<u>0.855 $\pm$ 0.029</u>	0.762 $\pm$ 0.022
GIN	1.678 $\pm$ 0.494	0.792 $\pm$ 0.097	0.716 $\pm$ 0.073	2.957 $\pm$ 0.696	0.990 $\pm$ 0.057	0.770 $\pm$ 0.021
EMGNN	0.936 $\pm$ 0.162	0.606 $\pm$ 0.041	0.639 $\pm$ 0.028	2.189 $\pm$ 0.128	0.834 $\pm$ 0.027	<u>0.743 $\pm$ 0.013</u>

S. H. Lee, F. Ji, K. Xia, and W. P. Tay, "Graph neural networks with a distribution of parametrized graphs," in Proc. International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024

<https://moleculenet.org/datasets-1>

Motivation: My Research

Project with **Continental Automotive**: AV localization using camera - detection of landmarks, matching, Kullback-Leibler divergence.



R. She, Q. Kang, S. Wang, W. P. Tay, Y. L. Guan, D. N. Navarro, and A. Hartmannsgruber, "Image patch-matching with graph-based learning in street scenes," IEEE Transactions on Image Processing, vol. 32, pp. 3465 – 3480, 2023

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Probabilistic Perspective

- What is a random variable X ?
 - ▶ X gives random values with certain probabilities. $p_X(x)$ is the “chance” or our “belief” that $X = x$.
 - ▶ Depending on whether X is discrete or continuous, $p_X(x)$ is called the **probability mass function (pmf)** or **probability density function (pdf)**.
 - ▶ In this course, we always assume either discrete random variable (pmf) or continuous random variable (pdf). In our discussion, we will just talk about pdf.
- Supervised learning: output Y may depend on factors other than given X . Treat the other factors as “random” effects. **Model input and output as a pair of random elements (X, Y) with joint pdf $p_{X,Y}(x, y)$.**
- Unsupervised learning: data X is generated by a “random” process. **Model data as a random element X with pdf $p_X(x)$.**

Examples of Famous Pdfs

- $X \sim \text{Bern}(\theta)$ has pmf $p_X(1) = \theta$, $p_X(0) = 1 - \theta$.
- Instead of the generic notation $p_X(x)$, we also write its pmf as

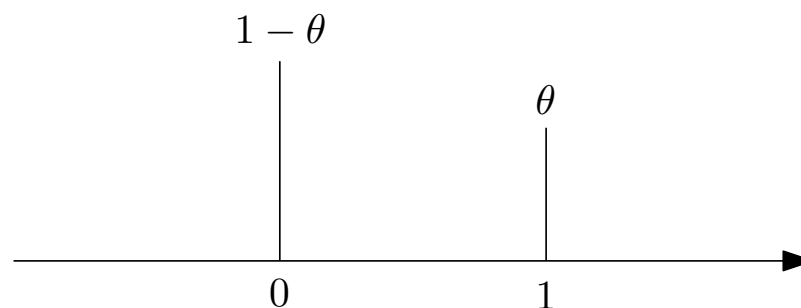
$$\begin{aligned} \text{Bern}(x | \theta) &= \begin{cases} \theta & \text{if } x = 1, \\ 1 - \theta & \text{if } x = 0, \end{cases} \\ &= \theta^x (1 - \theta)^{1-x} \\ &= \theta^{\mathbf{1}\{x=1\}} (1 - \theta)^{\mathbf{1}\{x=0\}}, \end{aligned}$$

where $\mathbf{1}\{A\} = 1$ if A is true and $\mathbf{1}\{A\} = 0$ if A is false.

- E.g.:

$$\text{Bern}(0 | 0.4) = 0.6 = 0.4^0 \cdot 0.6^1$$

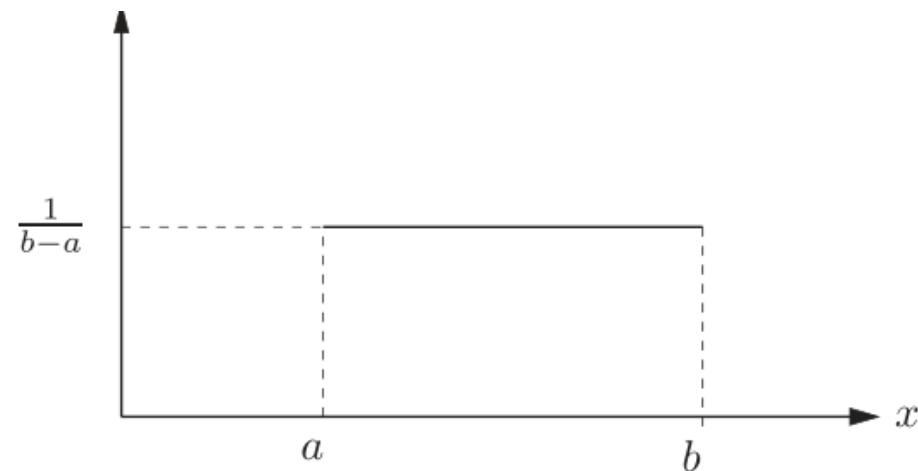
$$\text{Bern}(1 | 0.4) = 0.4 = 0.4^1 \cdot 0.6^0.$$



Examples of Famous Pdfs

- $X \sim \text{Unif}(a, b)$ has pdf

$$\text{Unif}(x | a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

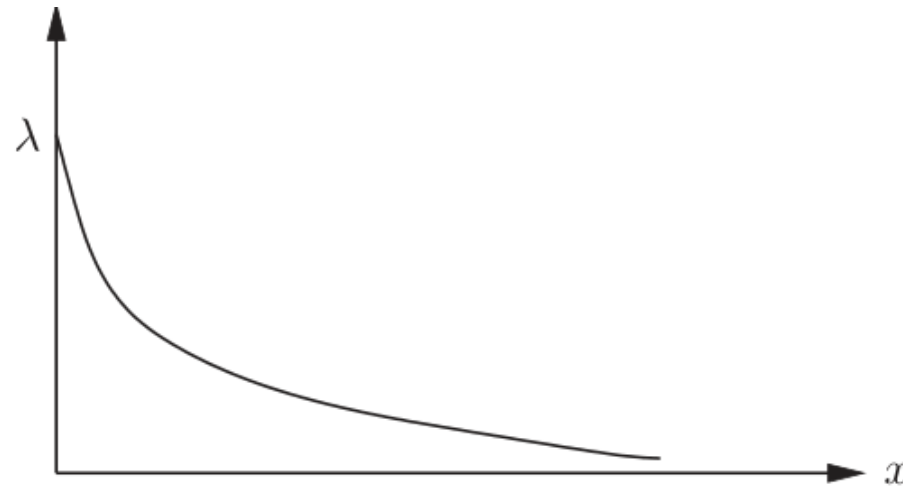


- E.g.: $\text{Unif}(x | 0, 1) = \frac{1}{1-0} = 1$ for $0 \leq x \leq 1$ and 0 otherwise.

Examples of Famous Pdfs

- $X \sim \text{Exp}(\lambda)$ has pdf

$$\text{Exp}(x | \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$



- E.g.: $\text{Exp}(x | 0.1) = 0.1e^{-0.1x}$ for $x \geq 0$ and 0 otherwise.

Probability Basics

Notation: we are lazy and often just write $p(x)$ for $p_X(x)$ and $p(x, y)$ for $p_{X,Y}(x, y)$. I.e., in most situations, we think of x as the realization of a random variable X .

- Joint pdf: $\mathbb{P}((X, Y) \in A) = \int_A p(x, y) \, dx \, dy$
- Marginal pdf: $p(x) = \int_{-\infty}^{\infty} p(x, y) \, dy$
- Conditional pdf: $p(y \mid x) = \frac{p(x, y)}{p(x)}$
- $p(x, y) = p(x)p(y \mid x)$
- **Bayes' rule:** $p(x \mid y) = \frac{p(x, y)}{p(y)} = \frac{p(y \mid x)p(x)}{p(y)}$
- If X and Y are **independent**, then $p(x, y) = p(x)p(y)$.

Dataset $\mathcal{D} = \{x_1, \dots, x_n\}$ independent and identically distributed (i.i.d.).

- Joint probability or likelihood: $p(\mathcal{D}) = \prod_{i=1}^n p(x_i)$
- Log likelihood: $\log p(\mathcal{D}) = \sum_{i=1}^n \log p(x_i)$

Probability Basics

Expectation

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xp(x) \, dx, \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)p(x) \, dx$$

$$\text{E.g., } p(x) = \int p(x | y)p(y) \, dy = \mathbb{E}_Y[p(x | Y)]$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[cX] = c\mathbb{E}[X]$$

$$\text{If } X \text{ and } Y \text{ are independent, } \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

$$\mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} xp(x | y) \, dx$$

$$\mathbb{E}[f(X)g(Y) | Y = y] = \mathbb{E}[f(X) | Y = y]g(y)$$

- Exercise: find the expectation of $\text{Bern}(\theta)$, $\text{Unif}(a, b)$, and $\text{Exp}(\lambda)$.

Probability Basics

$$\text{var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Uncorrelated: $\text{cov}(X, Y) = 0$. Note: independence $\implies$ uncorrelated, but not vice versa.

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2 \text{cov}(X, Y)$$

- Exercise: find the variance of $\text{Bern}(\theta)$, $\text{Unif}(a, b)$, and $\text{Exp}(\lambda)$.

Vector Variables

- $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \mathbb{E}[\mathbf{x}] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \\ \vdots \\ \mathbb{E}[x_n] \end{bmatrix}$

- If $\mathbf{A}$ is a deterministic matrix, $\mathbb{E}[\mathbf{Ax}] = \mathbf{A}\mathbb{E}[\mathbf{x}]$.

- Covariance: $\text{cov}(\mathbf{x}) = \Sigma_{\mathbf{xx}} = \mathbb{E}[(\mathbf{x} - \mathbb{E}\mathbf{x})(\mathbf{x} - \mathbb{E}\mathbf{x})^\top],$
 $\text{cov}(\mathbf{Ax}) = \mathbf{A} \text{cov}(\mathbf{x}) \mathbf{A}^\top$

- Cross-covariance: $\text{cov}(\mathbf{x}, \mathbf{y}) = \Sigma_{\mathbf{xy}} = \mathbb{E}[(\mathbf{x} - \mathbb{E}\mathbf{x})(\mathbf{y} - \mathbb{E}\mathbf{y})^\top]$

Vector Variables

- $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}$
- $\frac{\partial \mathbf{a}^\top \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{b}^\top$
- $\frac{\partial \det(\mathbf{X})}{\partial \mathbf{X}} = \det(\mathbf{X}) (\mathbf{X}^{-1})^\top$ (Note: $\det(\mathbf{X})$ is sometimes also written as $|\mathbf{X}|$.)
- $\frac{\partial \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{b}}{\partial \mathbf{X}} = -(\mathbf{X}^{-1})^\top \mathbf{a} \mathbf{b}^\top (\mathbf{X}^{-1})^\top$

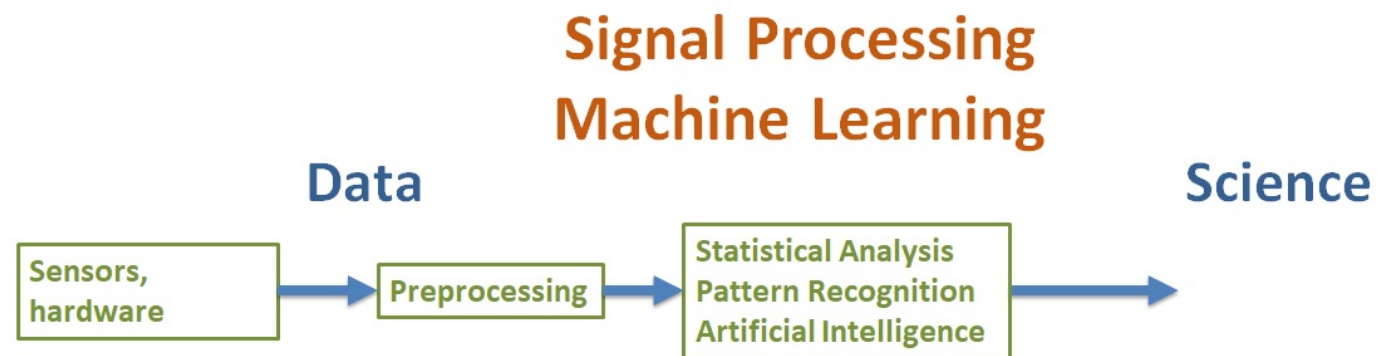
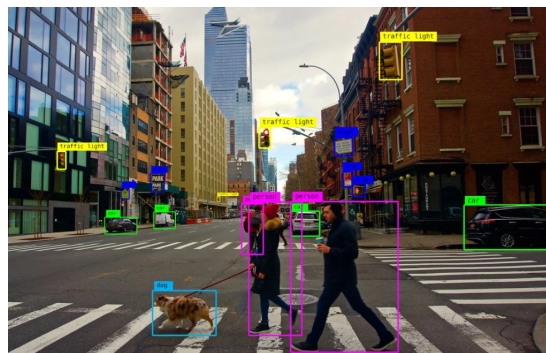
Matrix Cookbook

Parametric vs Non-parametric Models

- Parametric model: $p(x, y) = p(x, y \mid \theta)$ or $p(x) = p(x \mid \theta)$ has a fixed number of parameters, usually belongs to a fixed family of distributions.
 - ▶ Faster to train (find “optimal” θ).
 - ▶ Stronger assumptions about the data distribution
- Non-parametric model: number of parameters grow with the amount of training data.
 - ▶ More flexible
 - ▶ Can be computationally intractable for large datasets

Parametric Models

- Consider the following AV data (image, sound, Lidar, radar, etc.) as an example in the next few slides:



- We also consider cases where the raw data has been pre-processed and we have access only to features. Eg: data X is output from a detector for obstacle avoidance.

Parametric Models

- **Training a parametric model**: we assume a datum x is generated by a distribution $p(\cdot | \theta)$, where θ is a parameter. We want to **learn or infer** θ from observed data x :

$$\underbrace{p(\theta | x)}_{\text{posterior}} = \frac{p(x, \theta)}{p(x)} = \frac{p(x | \theta)p(\theta)}{p(x)} \propto \underbrace{p(x | \theta)}_{\text{likelihood}} \cdot \underbrace{p(\theta)}_{\text{prior}}$$

- WLOG, can ignore the constant $p(x)$ as it does not depend on θ .
 - If you know $p(x | \theta)p(\theta)$, you can easily recover $p(x) = \int p(x | \theta)p(\theta) d\theta$.
- Remark: If you know $p(x) \propto f(x)$ where $f(\cdot)$ is any non-negative function or *functional form*, then the pdf $p(x) = cf(x)$ is fixed and known with $c = \left(\int f(x) dx\right)^{-1}$.

Conjugate Distributions

$$\underbrace{p(\theta | x)}_{\text{posterior}} \propto \underbrace{p(x | \theta)}_{\text{likelihood}} \cdot \underbrace{p(\theta)}_{\text{prior}}$$

$$\log p(\theta | x) = \underbrace{\log p(x | \theta)}_{\text{log likelihood}} + \log p(\theta) + \text{const}$$

- If $p(\theta)$ and $p(\theta | x)$ have the **same form** (belongs to same distribution family), we say that they are *conjugate distributions*, and $p(\theta)$ is a *conjugate prior* for the likelihood $p(x | \theta)$.
- Why: Conjugate distributions allow for **analytical closed form** solutions, simple to implement, and
- Easy to interpret: the posterior is an “updated” version of the prior.
- Con: may not be flexible enough to capture complex data. **MCMC** methods are used in such cases.

Example: Bernoulli Distribution

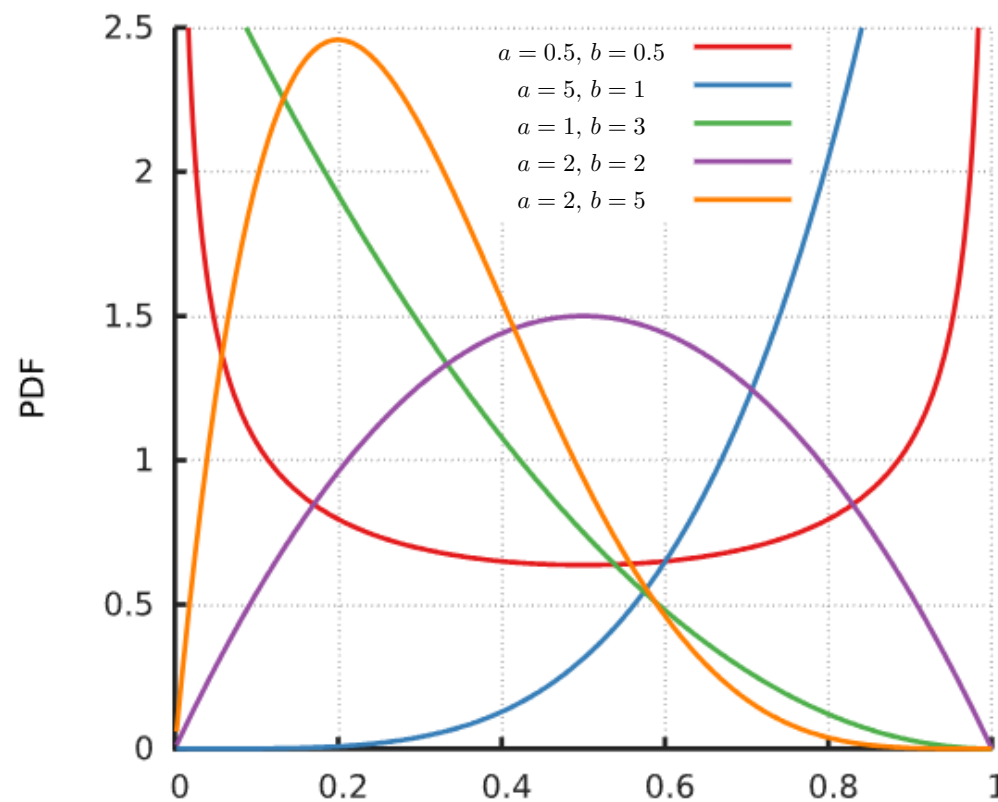
- $\text{Bern}(x | \theta)$, $\theta \geq 0$: $x \in \{0, 1\}$ with $\text{Bern}(1 | \theta) = \theta$.
- $\text{Bern}(x | \theta) = \theta^x (1 - \theta)^{1-x}$
 - ▶ AV: x indicates if an obstacle is detected. As the detection is imperfect, we can only say $X = 1$ with a certain probability θ .
 - ▶ θ depends on the sensor, the environment, etc., and is unknown *a priori*.

Beta

- **Prior model:** Beta ($\theta \mid a, b$), $a, b > 0$, $\theta \in [0, 1]$:

$$\text{Beta}(\theta \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1},$$

where $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$. The factor $\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$ simply ensures that $\int_{\mathbb{R}} p(\theta) d\theta = 1$.



Beta as Conjugate Prior

- Conjugate prior to $\text{Bern}(\cdot | \theta)$:

$$\begin{aligned}
 \overbrace{p(\theta | x)}^{\text{posterior}} &\propto \overbrace{p(x | \theta)}^{\text{likelihood}} \overbrace{p(\theta)}^{\text{prior}} \\
 &\propto \overbrace{\text{Bern}(x | \theta)}^{\text{likelihood}} \overbrace{\text{Beta}(\theta | a, b)}^{\text{prior}} \\
 &\propto \theta^x (1 - \theta)^{1-x} \theta^{a-1} (1 - \theta)^{b-1} \\
 &\propto \theta^{x+a-1} (1 - \theta)^{1-x+b-1}.
 \end{aligned}$$

- $\therefore p(\theta | x) = \text{Beta}(\theta | x + a, 1 - x + b)$
- Posterior is also a Beta distribution with updated parameters. I.e., the **Beta distribution is a conjugate prior for the Bernoulli distribution.**

I.I.D. Dataset

- Given dataset $\mathcal{D} = \{x_1, \dots, x_n\}$ i.i.d. $\sim \text{Bern}(\cdot | \theta)$.

$$\begin{aligned}
 p(\theta | \mathcal{D}) &\propto p(\mathcal{D} | \theta) p(\theta) \\
 &= \prod_{i=1}^n p(x_i | \theta) \cdot p(\theta) \\
 &\propto \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \cdot \theta^{a-1} (1 - \theta)^{b-1} \\
 &\propto \theta^{\sum_{i=1}^n x_i + a - 1} (1 - \theta)^{n - \sum_{i=1}^n x_i + b - 1}.
 \end{aligned}$$

- $\therefore p(\theta | \mathcal{D}) = \text{Beta}(\theta | \sum_{i=1}^n x_i + a, n - \sum_{i=1}^n x_i + b)$.
- Log likelihood:

$$\begin{aligned}
 \log p(\mathcal{D} | \theta) &= \log \prod_{i=1}^n p(x_i | \theta) = \sum_{i=1}^n \log p(x_i | \theta) \\
 &= \sum_{i=1}^n (x_i \log \theta + (1 - x_i) \log(1 - \theta)).
 \end{aligned}$$

Example: Categorical Distribution

- $\text{Cat}(\cdot | \theta_1, \dots, \theta_K)$, $\theta_k \geq 0$, $\sum_{k=1}^K \theta_k = 1$: $\text{Cat}(x | \theta_1, \dots, \theta_K) = \theta_x$.
 - ▶ AV: X is detection of {pedestrian, bicycle, car, truck, lamp-post}.
- Suppose $X_i \sim \text{Cat}(\cdot | \theta_1, \dots, \theta_K)$ i.i.d., $i = 1, 2, \dots, n$. Let $\mathcal{D} = \{X_1, \dots, X_n\}$. Joint pdf of $X_1, \dots, X_n$ is

$$p(\mathcal{D}) = \prod_{i=1}^n \text{Cat}(x_i | \theta_1, \dots, \theta_K) = \prod_{i=1}^n \prod_{k=1}^K \theta_k^{\mathbf{1}\{x_i=k\}} = \prod_{k=1}^K \theta_k^{N_k},$$

where $N_k = \sum_{i=1}^n \mathbf{1}\{x_i = k\}$.

- ▶ AV: X_i is detection of {pedestrian, bicycle, car, truck, lamp-post} in box i , n bounding boxes in total.

Example: Dirichlet (Supplementary)

- For $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)$ such that $\theta_k \in [0, 1]$, $\forall k$, and $\sum_{k=1}^K \theta_k = 1$,

$$\text{Dir}(\boldsymbol{\theta} \mid \alpha_1, \dots, \alpha_K) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_K)} \prod_{k=1}^K \theta_k^{\alpha_k - 1},$$

- The space $\left\{ \boldsymbol{\theta} : \theta_k \in [0, 1], \sum_{k=1}^K \theta_k = 1 \right\}$ is the space of *all* categorical distributions over K categories.
- $\text{Dir}(\boldsymbol{\theta} \mid \alpha_1, \dots, \alpha_K)$ is a *distribution over the space of categorical distributions* (with K categories). Beta distribution is $K = 2$.
- Conjugate prior to $\text{Cat}(\cdot \mid \boldsymbol{\theta})$:

$$\overbrace{p(\boldsymbol{\theta} \mid \boldsymbol{s})}^{\text{posterior}} = \overbrace{\text{Cat}(\boldsymbol{x} \mid \boldsymbol{\theta})}^{\text{likelihood}} \overbrace{\text{Dir}(\boldsymbol{\theta} \mid \boldsymbol{\alpha})}^{\text{prior}} \propto \prod_{k=1}^K \theta_k^{\mathbf{1}\{x=k\}} \prod_{k=1}^K \theta_k^{\alpha_k - 1} \propto \prod_{k=1}^K \theta_k^{\mathbf{1}\{x=k\} + \alpha_k - 1}$$

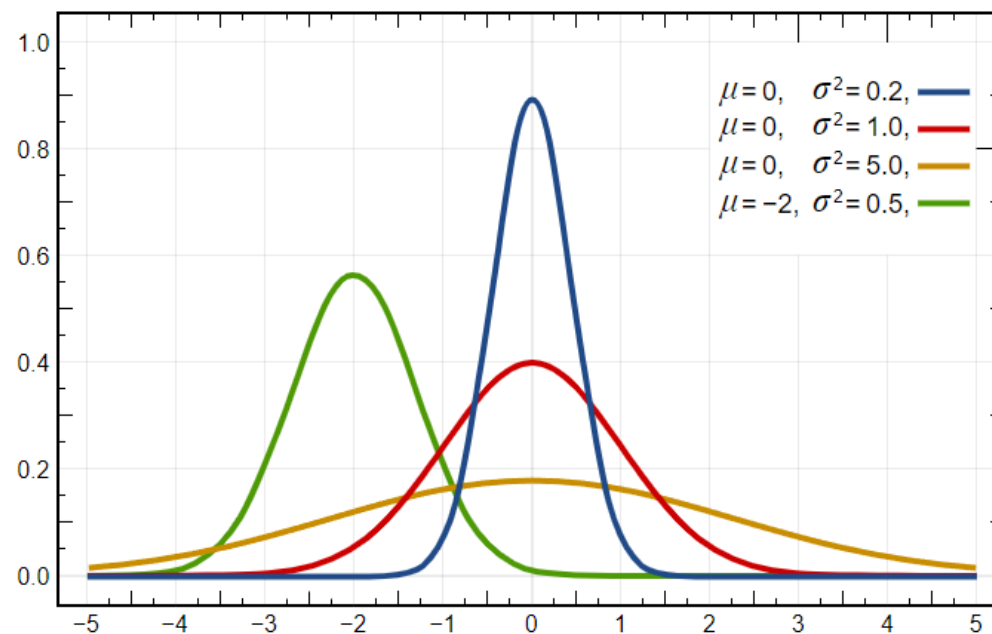
$$\therefore p(\boldsymbol{\theta} \mid \boldsymbol{s}) = \text{Dir}(\boldsymbol{\theta} \mid \boldsymbol{s} + \boldsymbol{\alpha}),$$

where $\boldsymbol{s} = (\mathbf{1}\{x = 1\}, \dots, \mathbf{1}\{x = K\})$, $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$.

Example: Gaussian or Normal Distribution

- Gaussian or normal distribution: $x \in \mathbb{R}$

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



- ▶ $\mathcal{N}(x \mid 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ is called the **standard normal distribution**.
- ▶ AV: X is a radar range measurement from an obstacle at a distance μ .

Example: Gaussian or Normal Distribution

Some useful properties:

- If $X \sim \mathcal{N}(\mu, \sigma^2)$, then
 - ▶ $aX \sim \mathcal{N}(a\mu, a^2\sigma^2)$ for any constant $a \in \mathbb{R}$
 - ▶ $X + c \sim \mathcal{N}(\mu + c, \sigma^2)$
 - ▶ Let $Z \sim \mathcal{N}(0, 1)$ (standard normal). Then $X = \sigma Z + \mu \sim \mathcal{N}(\mu, \sigma^2)$.
- If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y \sim \mathcal{N}(\xi, \nu^2)$ are independent, then

$$X + Y \sim \mathcal{N}(\mu + \xi, \sigma^2 + \nu^2).$$

Jointly Gaussian Distribution

- Definition: $\mathbf{X} = [X_1, \dots, X_n]^\top$ is jointly Gaussian or multivariate Gaussian if for any $\mathbf{a} = [a_1, \dots, a_n]^\top \in \mathbb{R}^n$, the random variable

$$\mathbf{a}^\top \mathbf{X} = \sum_{i=1}^n a_i X_i$$

is Gaussian.

- Any linear combination of the components of $\mathbf{X}$ is Gaussian.

Jointly Gaussian Distribution

- Let $Z_1, \dots, Z_K \sim \mathcal{N}(\cdot | 0, 1)$ i.i.d. They are jointly Gaussian. Why?
- Let

$$\mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu}, \quad \mathbf{Z} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_K \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix},$$

where $\mathbf{A} = [a_{ij}]_{i=1,j=1}^{n,K}$ is a given array of real numbers and $\boldsymbol{\mu}$ is a vector of real numbers.

- Then, $\mathbf{X}$ is jointly Gaussian with mean and covariance given by

$$\mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbf{A}\mathbf{Z} + \boldsymbol{\mu}] = \mathbf{A}\mathbb{E}\mathbf{Z} + \boldsymbol{\mu} = \boldsymbol{\mu}$$

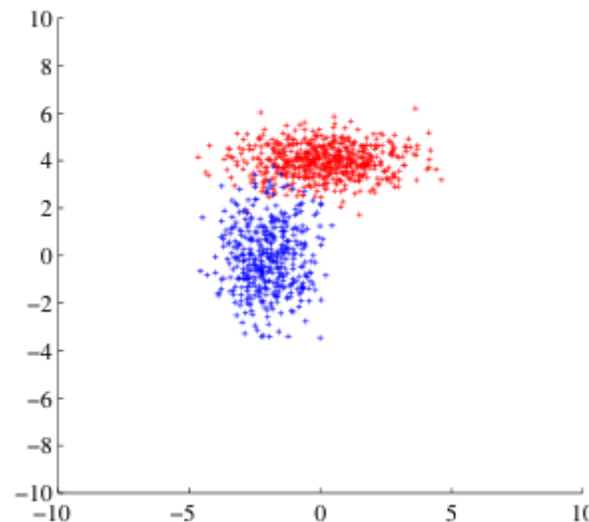
$$\text{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] = \mathbb{E}[\mathbf{A}\mathbf{Z}\mathbf{Z}^\top\mathbf{A}^\top] = \mathbf{A}\mathbb{E}[\mathbf{Z}\mathbf{Z}^\top]\mathbf{A}^\top = \mathbf{A}\mathbf{A}^\top.$$

Jointly Gaussian Distribution

-

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{K/2} (\det \boldsymbol{\Sigma})^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (1.1)$$

- AV: Suppose $\mathbf{X}$ consists of 2D LiDAR information.



Joint Gaussian Distribution Examples

Two-dimensional jointly Gaussian $\mathbf{X} = \begin{bmatrix} X \\ Y \end{bmatrix}$ with

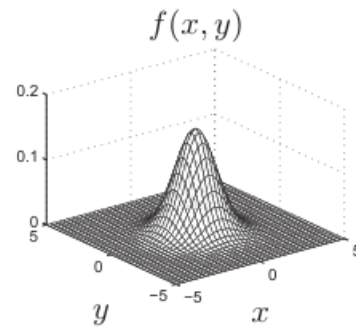
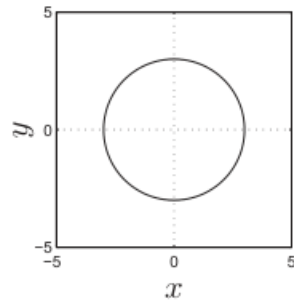
$$\boldsymbol{\mu} = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_X^2 & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \sigma_Y^2 \end{bmatrix}$$

Correlation coefficient:

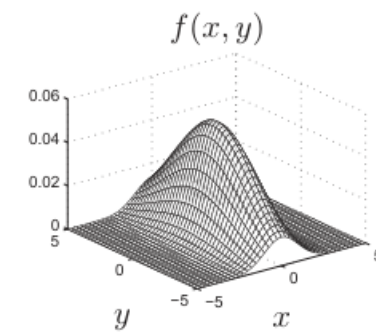
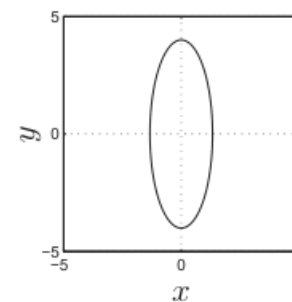
$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Joint Gaussian Distribution Examples

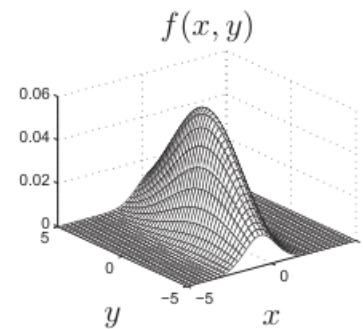
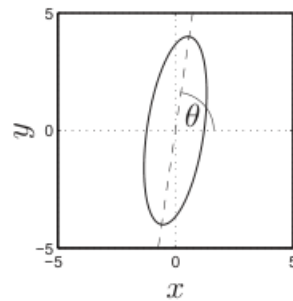
$$\sigma_X = 1, \sigma_Y = 1, \rho_{X,Y} = 0:$$



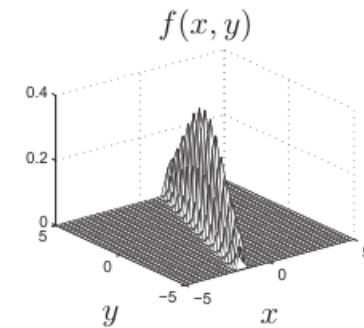
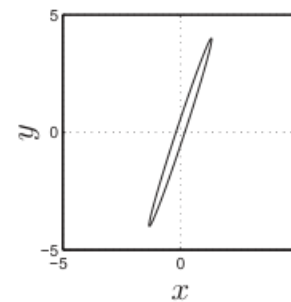
$$\sigma_X = 1, \sigma_Y = 3, \rho_{X,Y} = 0:$$



$$\sigma_X = 1, \sigma_Y = 3, \rho_{X,Y} = 0.4:$$



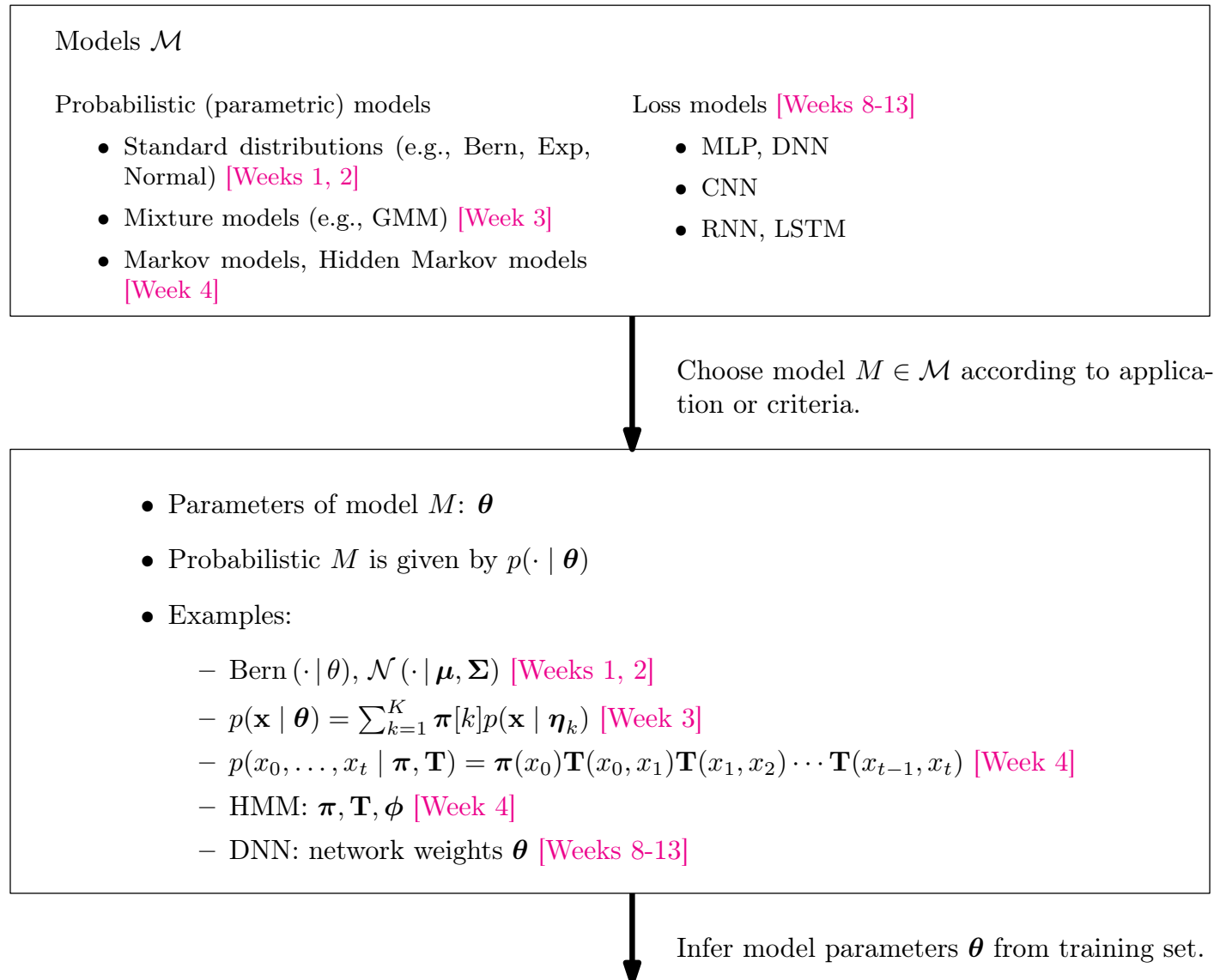
$$\sigma_X = 1, \sigma_Y = 3, \rho_{X,Y} = 0.99:$$



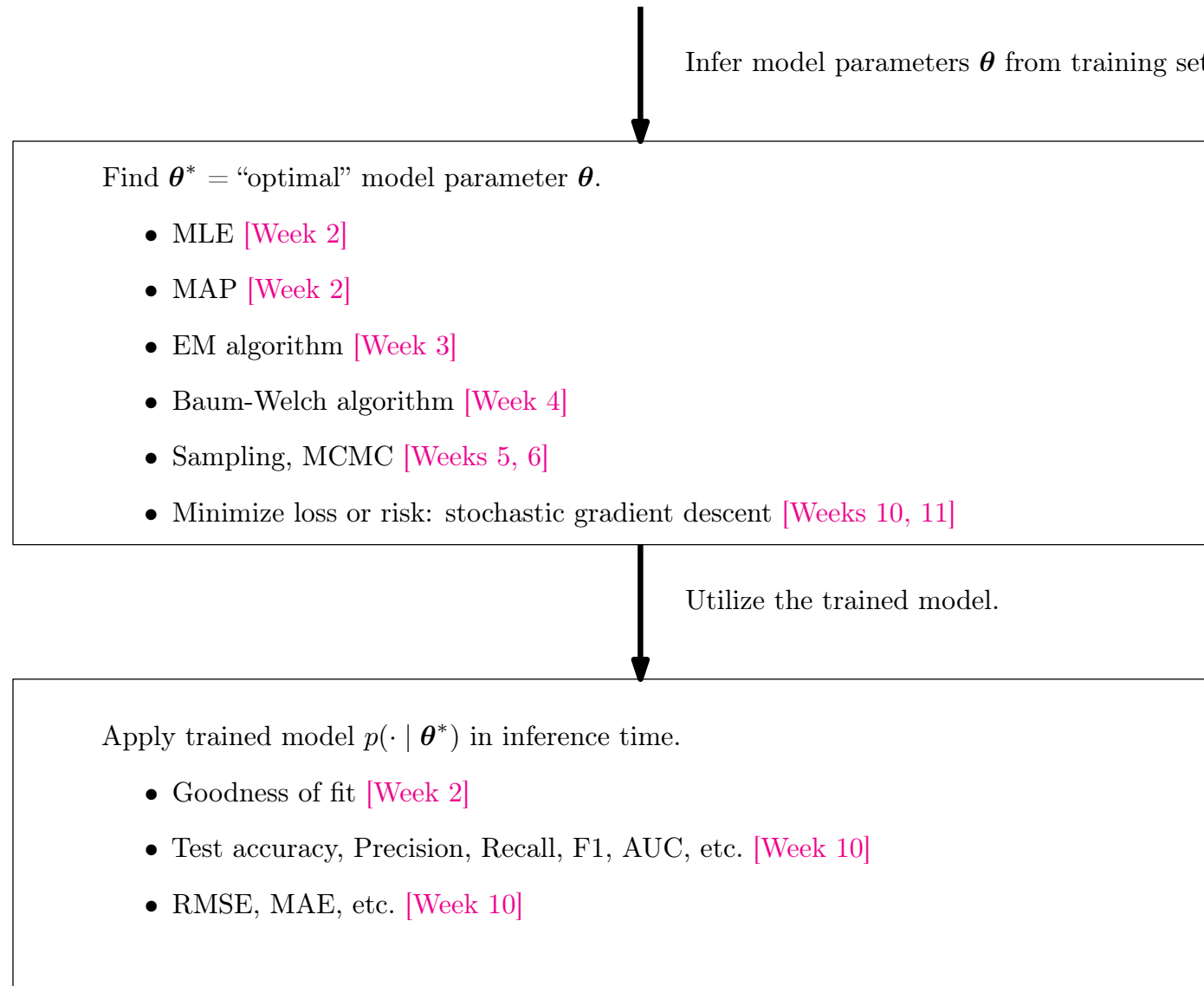
Summary

- Some basic probability theory.
 - ▶ Pdfs and their properties
 - ▶ Expectation and variance
 - ▶ Vector variables
 - ▶ Conjugate distributions
- Some standard distributions.
 - ▶ Unif, Exp
 - ▶ Bern, Beta
 - ▶ Cat, Dir (Supplementary)
 - ▶ Normal/Gaussian

The Path Ahead



The Path Ahead



Outline

- 1 Course Admin
- 2 Overview
- 3 Probability Review
- 4 Some Practice Problems

Some Practice Problems

1. Let $X \sim \text{Bin}(\cdot | \theta, n)$ be distributed according to the binomial distribution, then $X = \sum_{i=1}^n Y_i$ where $Y_i \sim \text{Bern}(\cdot | \theta)$ i.i.d. Find the mean and variance of X .

Solution:

$$\mathbb{E}X = \sum_{i=1}^n \mathbb{E}Y_i = n\theta,$$

$$\text{var } X = \sum_{i=1}^n \text{var } Y_i = \sum_{i=1}^n (\theta(1 - \theta)^2 + (1 - \theta)\theta^2) = n\theta(1 - \theta).$$

2. Determine if the following statements are True or False.

(a) If $\log p(x) = -2x^3 - 4x^2 + x + \text{const}$, then $p(x)$ is a Gaussian distribution.

(b) If X and Y are i.i.d. $\mathcal{N}(5, 2)$ random variables, then $\mathbf{Z} = \begin{bmatrix} X \\ Y \end{bmatrix}$ is jointly

Gaussian.

Some Practice Problems (cont.)

Solution:

- (a) False: Not quadratic in x in its exponent.
- (b) True: Any linear combination of X and Y is Gaussian.

3. The dataset $\mathcal{D} = \{x_1 = 0.3, x_2 = 0.6, x_3 = 0.8\}$ is generated i.i.d. from $\text{Exp}(\cdot | \lambda)$. Write down the joint probability or likelihood of $\mathcal{D}$. Write down the log likelihood.

Solution:

$$\begin{aligned}
 p(\mathcal{D} | \lambda) &= \prod_{i=1}^3 p(x_i | \lambda) \\
 &= \prod_{i=1}^3 \lambda e^{-\lambda x_i} \\
 &= \lambda^3 e^{-\lambda(0.3+0.6+0.8)} \\
 &= \lambda^3 e^{-1.7\lambda}.
 \end{aligned}$$

$$\log p(\mathcal{D} | \lambda) = 3 \log \lambda - 1.7\lambda.$$

Some Practice Problems (cont.)

4. You are given the training set $\mathcal{D} = \{x_1, \dots, x_n\}$, where $x_i \sim \text{Bern}(\cdot | \theta)$ i.i.d. for all i . Write down the log likelihood $\log p(\mathcal{D} | \theta)$.

Solution:

$$\begin{aligned} p(\mathcal{D} | \theta) &= \prod_{i=1}^n p(x_i | \theta) \\ &= \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \\ &= \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i} \\ \log p(\mathcal{D} | \theta) &= \left(\sum_{i=1}^n x_i \right) \log \theta + \left(n - \sum_{i=1}^n x_i \right) \log(1 - \theta). \end{aligned}$$

Some Practice Problems (cont.)

5. You are given the training set $\mathcal{D} = \{x_1, \dots, x_n\}$, where $x_i \sim \mathcal{N}(\cdot \mid \mu, \sigma^2)$ i.i.d. for all i . Write down the log likelihood $\log p(\mathcal{D} \mid \mu, \sigma^2)$.

Solution:

$$\begin{aligned}\log p(\mathcal{D} \mid \mu, \sigma^2) &= \sum_{i=1}^n \log \mathcal{N}(x_i \mid \mu, \sigma^2) \\ &= \sum_{i=1}^n \left(-\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (x_i - \mu)^2 \right) \\ &= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.\end{aligned}$$

Some Practice Problems (cont.)

6. Let $\mathbf{X}$ be a random vector. Show that $\text{cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top$.

Solution:

$$\begin{aligned}\text{cov}(\mathbf{X}) &= \mathbb{E}[(\mathbf{X} - \mathbb{E}\mathbf{X})(\mathbf{X} - \mathbb{E}\mathbf{X})^\top] \\ &= \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}\mathbf{X}\mathbb{E}\mathbf{X}^\top - \mathbb{E}\mathbf{X}\mathbb{E}\mathbf{X}^\top + \mathbb{E}\mathbf{X}\mathbb{E}\mathbf{X}^\top \\ &= \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top.\end{aligned}$$